

rank-order correlations were used to assess relationships between pain and diet variables, with alpha set at 0.05 (uncorrected).

Results: The study sample (n=46) primarily consisted of males (75%), with about half (54%) with tetraplegia, a mean age of 40.7 ± 11.5 y, a mean weight of 79.3 ± 18.9 kg, and a mean HEI of 55.3 ± 15.0 . HEI was negatively correlated with NPSI ($r = -0.322$; $p = 0.042$) but not associated with ratings of average pain intensity or unpleasantness ($p > 0.05$). Average energy intake adjusted by body weight was positively correlated with average pain intensity rating ($r = 0.311$; $p = 0.042$) and NPSI scores ($r = 0.294$, $p = 0.047$). Vitamin D ($r = -0.328$; $p = 0.039$), magnesium ($r = -0.323$; $p = 0.042$), and potassium ($r = -0.334$; $p = 0.035$) were all negatively correlated with ratings of average unpleasantness (all, $p \leq 0.042$).

Conclusions: Our preliminary analysis suggests that less healthy eating patterns, including excess energy intake, are related to greater levels of pain in individuals with SCI. Poor diet after SCI may instigate an inflammatory environment, increasing pain susceptibility.

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P09-030-25 Vitamin B6 and Inositol Potentially Inhibit Neuronal Excitotoxicity Associated With Ischemic Stroke – Molecular Docking Investigation

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Objectives: The objective of the study is to investigate Lutein, Vitamin B6, Inositol, and Choline as potential inhibitors for N-methyl-D-aspartate (NMDA) receptor excitotoxicity following Ischemic Stroke using computational molecular docking methods.

Methods: Molecular docking modeling and analysis was performed using AutoDock Tool 4.2 version software to model the binding interaction and calculate the binding affinity between the NMDA receptor subunit N2B and the following micronutrient ligands: Lutein, Vitamin B6, Inositol, and Choline. The positive control ligand for the study was Glutamate, which is known to bind to the N2B subunit of the NMDA receptor.

Results: Among the 4 micronutrient binding interactions with the NMDA receptor, Vitamin B6 and Inositol demonstrated a strong binding interaction with a mean binding affinity of -6.46 kcal/mol and -6.34 kcal/mol respectively in comparison to the control, Glutamate, which had a mean binding affinity of -3.73 kcal/mol. Lutein and Choline interaction with the NMDA receptor had a low binding affinity and weak strength in binding interaction in comparison to other ligands with a mean binding affinity of -2.92 kcal/mol and -3.95 kcal/mol respectively.

Conclusions: The molecular docking findings identified Vitamin B6 and Inositol as potential excitotoxic-inhibitors of the NMDA receptor NB2 subunit. These findings suggest the

neuroprotective effects of Vitamin B6 and Inositol may be potential therapeutic micronutrients to address excitotoxicity following cerebrovascular accidents and prevent the severity of neuronal damage and disability caused by Ischemic Stroke.

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P09-031-25 Resistant Starch Supplementation Promotes Brain Health in Estrogen Deficient Mice by Modulating Gut-Derived Enzyme and Metabolites

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Objectives: This study investigated the effects of two types of resistant starch (RS), retrograded (RS3) and chemically modified (RS4), on neuronal genes, gut microbial β -glucuronidase (*gmGUS*) activity, and metabolites affecting brain health in a mouse model of estrogen deficiency.

Methods: 3-mo-old female C57BL/6 mice were injected (ip) daily for 30 d with 4-vinylcyclohexene diepoxide (VCD, 160 mg/kg bw in sesame oil), or vehicle (Veh) to induce estrogen deficiency. Mice were then divided into three dietary groups (n=15/group) in a 2x3 factorial design: control (AIN-93M), 5% (wt/wt) RS3 or RS4 for 16 wks. Fecal *gmGUS* activity, involved in deconjugating estrogen and neurotransmitters, was measured. Serum tryptophan and its metabolites (kynurenine, 3-hydroxykynurenine, quinolinic and kynurenic acids), important for immune and nervous system function, were analyzed using liquid chromatography-tandem mass spectrometry (LC-MS/MS). Hippocampal RNA-seq was performed to assess genes important in brain health. Data were analyzed using two-way ANOVA with estrogen status (VCD/Veh) and diet as factors, followed by *Tukey's post-hoc* tests when $P < 0.05$.

Results: Estrogen deficiency tended ($P = 0.056$) to decrease relative brain weight, while there was a main effect of diet with RS3 ($P = 0.022$) and RS4 ($P = 0.069$) decreasing brain weight compared to the control. Estrogen status had no significant effect on *gmGUS* activity; however, there was a main effect of diet, with RS4 increasing ($P = 0.048$) enzyme activity compared to the control. VCD significantly reduced serum kynurenine and 3-hydroxykynurenine concentrations ($P_{adj} < 0.005$), while diet tended ($P = 0.065$) to affect only kynurenine. RNA-seq data revealed differential expression of genes involved in cognitive function. In the Veh group, both RS significantly increased *Pax7* expression. In the VCD group, RS3 upregulated *Thoc2* and *Paxbp1*, while both RS treatments increased *Pura* expression. In estrogen-deficient mice, *C1ptm1* and *Vdac1*, genes linked to Alzheimer's disease, were downregulated by RS3 and RS4, respectively.

Conclusions: This study provides evidence that VCD has a negative effect on tryptophan metabolites, while RS3 and RS4